Position, distance & displacement

Distance = add all path lengths.

Displacement = final position - initial position.

Math Skills

1. 12 + 5

2. 9 - (-6)

3. 18 - 18

Practice

1. The hallway delivery

A student carries a box 12 m east from the classroom to the office, then walks 5 m west to a supply closet. What total distance does the student travel?

2. Across the reference line

On a straight east-west trail, the water fountain is position 0 m. A walker starts at -6 m and finishes at +9 m. Take east as positive. Find the displacement, including direction.

3. One complete lap

A runner completes one full lap of a circular track and stops exactly where she started. Is her distance zero, her displacement zero, both, or neither? Explain using the start and finish.

4. Seated on the train

A passenger stays seated while a train moves east at 10 m/s. Describe the passenger's motion relative to the train, and then relative to the station platform. Why do the descriptions differ?

Speed & velocity

Average speed = total distance ÷ total elapsed time.

Distance = speed × time. Velocity is speed with direction.

Math Skills

1. 180 ÷ 120

2. 4 × 45

3. Solve: 3t = 90.

Practice

5. Walking the trail

A student walks 180 m along a straight trail in 120 s. What is the student's average speed? Include the correct unit.

6. Cycling along the path

A cyclist rides straight east at a steady 4 m/s for 45 s. How far does the cyclist travel?

7. Jogging in opposite directions

Two joggers pass a fountain. One moves east at 2 m/s; the other moves west at 2 m/s. Do they have the same speed? Do they have the same velocity? Explain.

8. The dog stops to sniff

A dog travels along a path at 2 m/s for 10 s, stops for 10 s, then continues at 2 m/s for 10 s. Is its average speed for all 30 s equal to 2 m/s? Explain why the stop matters.

Acceleration

Straight-line motion: $\mathbf{a} = (v_f - v_i) \div \Delta t$ $\mathbf{v_f} = v_i + a \times \Delta t$ $\mathbf{\Delta t}$ = final time - initial time

v_i = initial velocity; v_f = final velocity. Right/east is positive.

Math Skills

1. 8 - 3

2. (14 - 4) ÷ 5

3. 2 + (0.5 × 4)

Practice

9. The bus pulls away

A bus travels east at 4 m/s when a stopwatch reads 3 s. At 8 s, it travels east at 14 m/s. What is its average acceleration during this interval?

10. The runner speeds up

A runner moves east at 2 m/s, then accelerates at 0.5 m/s² for 4 s while continuing east. What is the runner's final velocity?

11. Backing up faster

A small remote-control car moves west. Its velocity changes from -1 m/s to -3 m/s in 2 s. Is it speeding up or slowing down? What does the negative sign tell you?

12. Earth in orbit

Model Earth's orbit around the Sun as a circle traveled at constant speed. Is Earth accelerating in this model? Explain what changes and what force keeps its path curved.

Reading motion graphs

Slope = change in vertical value ÷ change in time.

Distance-time slope gives average speed for that interval. For straight motion in the positive direction, speed-time slope gives acceleration.

Math Skills

1. (30 - 0) ÷ (30 - 0)

2. (8 - 2) ÷ (6 - 0)

3. Compare: 8/10, 16/10, 4/10. Which is greatest?

Practice

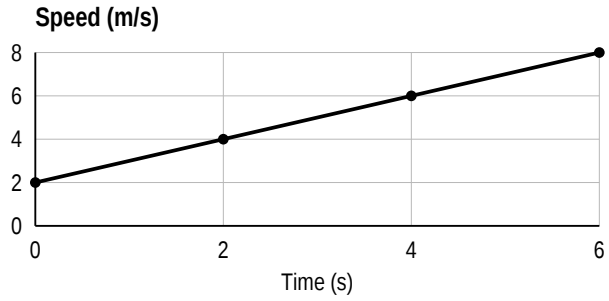
13. A pause on the nature walk

The table shows a student's total distance along a trail. The student walks forward, pauses, then walks forward again. What is the average speed over the entire 30 s, including the pause?

Time (s)	0	10	20	30
Total distance (m)	0	12	12	30

14. Reading a scooter's speed graph

The scooter's speed increases steadily as shown. It travels east throughout. Use two points to find its acceleration from 0 s to 6 s.



15. Which interval is fastest?

A walker's distance-time data are shown below. Which 10 s interval has the greatest average speed? Explain how the distances, or the steepness of a graph, support your choice.

Time (s)	0	10	20	30
Total distance (m)	0	8	24	28

16. What a speed graph cannot tell you

Two cyclists each ride steadily at 3 m/s for 6 s. One rides east and one rides west. Would their speed-time graphs differ? Could those graphs alone tell you their directions?

Gravity, planets & friction

Weight = mass × g. Earth: g = 10 m/s². Mars: g = 3.7 m/s².

Mass stays the same when location changes; weight depends on g.
Static friction adjusts to the push, up to its maximum.

Math Skills

1. 20 × 3.7

2. Solve: f - 35 = 0.

3. 150 × 3.7

Practice

17. A science instrument on Mars

A science instrument has a mass of 20 kg. What would its weight be on the surface of Mars? Use Mars's gravitational acceleration, g = 3.7 m/s².

18. Keeping the box moving steadily

A student pushes a plastic equipment box across a smooth floor at a steady speed. The push is 35 N to the right. What is the sliding friction force on the box? Give its size and direction.

19. The same astronaut on another planet

An astronaut and suit have a combined mass of 150 kg on Earth. If they were on Mars, where surface gravity is weaker, would their mass change? Would their weight change? Explain.

20. A crate that will not move

A student pulls a crate right with 10 N, but it stays at rest. Static friction can reach a maximum of 18 N. What is the actual friction force now? Explain why it is not automatically 18 N.

Vertical net forces & Newton's first law

$\mathbf{F_{net}}$ = upward forces - downward forces. Choose upward as positive.
$\mathbf{F_{net} = 0}$ means constant velocity, which can include rest.

Math Skills

1. 15,000 - 12,000

2. 9,800 + (-10,000)

3. 12,000 - 12,000

Practice

21. Forces on a helicopter

A helicopter experiences an upward lifting force of 15,000 N and a downward weight of 12,000 N. Ignore other vertical forces. What is the vertical net force? Include its direction.

22. Forces on a hot-air balloon

A hot-air balloon has an upward buoyant force of 9,800 N and a total weight of 10,000 N. Ignore air resistance. Take upward as positive. Find the vertical net force, including its sign and direction.

23. Balanced forces in flight

An airplane flies straight and level at constant speed. Upward lift balances its weight, and forward thrust balances drag. Must the airplane stop because the forces balance? Explain what zero net force means.

24. After the hockey stick lets go

A hockey puck glides across very smooth ice after the stick is no longer touching it. Over a short interval, ignore the small friction and air resistance. What happens to its horizontal velocity? Is a continuing forward force required?

Newton's second law

$F_{net} = ma$ **$a = F_{net} \div m$** **$m = F_{net} \div a$**
Force: N Mass: kg Acceleration: m/s². For vertical motion, upward is positive.
For vertical motion: **F_{net}** = upward forces - weight; **weight = mg**.

Math Skills

1. (6 - 5) ÷ 0.50

2. Solve: 1.5m = 3.

3. 0.8 ÷ 0.4; 0.8 ÷ 0.8

Practice

25. A bird takes off

Just after leaving the ground, the air exerts an average upward force of 6 N on a 0.50 kg bird. Its weight is 5 N downward. Ignore other vertical forces. Find the bird's average vertical acceleration.

26. Finding a drone's mass

A drone accelerates upward at 1.5 m/s² when the net upward force on it is 3 N. What is the drone's total mass, including its battery?

27. Two model airplanes

Two model airplanes have masses of 0.4 kg and 0.8 kg. The same 0.8 N forward net force acts on each. Which has the greater acceleration? How does doubling the mass affect acceleration when net force stays the same?

28. Force through a bicycle turn

A cyclist makes a circular turn on level pavement at constant speed. In which direction must the net horizontal force point: forward, outward, or toward the center? Which contact force from the road can provide it?

Collisions, momentum & Newton's third law

Momentum: p = mv. Total system momentum: $\mathbf{p_{total} = m_1v_1 + m_2v_2}$.
Total momentum is conserved when external forces have negligible effect during collision.
A third-law force pair is equal and opposite and acts on two different objects.

Math Skills

1. (6 × 3) + (1.5 × 0) **2.** 0.10 × 2 **3.** 0.20 + 0

Practice

29. A bowling ball and a pin

A 6 kg bowling ball rolls forward at 3 m/s toward a stationary 1.5 kg pin. What is the total momentum of the ball-and-pin system just before impact?

30. Clay balls stick together

A 0.10 kg clay ball moving right at 2 m/s hits a stationary 0.10 kg clay ball on a smooth, level surface. They stick together. Ignore external horizontal forces during the brief collision. What is the total momentum of the two-ball system before and immediately after the collision?

31. Two marbles collide

A moving marble hits a stationary marble. During the collision, does the moving marble exert a larger force on the stationary one, or are the forces equal in size? Explain the directions and which object each force acts on.

32. A curling-stone collision

A curling stone hits another stone at rest. The first slows down and the second starts moving. Ignore external horizontal forces during the brief collision. Has momentum disappeared? Explain what happens to each stone's momentum and their combined momentum.